Carla Becker

Leadership & Management

- **People Leadership**
  3+ years managing technical teams (up to 10 direct reports); 1:1 coaching and mentorship; hiring, onboarding, and training; performance feedback and career development; navigating ambiguity and maintaining team morale through change.
- **Program Management**
  Technical program management, roadmap definition, early-stage project ramp-up; cross-functional coordination across R&D, software, hardware, and operations; vendor management, SOWs, budgeting, and delivery against milestones.
- **Technical Background**
  ML model evaluation; simulation-ML hybrid systems; Python, SQL, Flask, Docker/Kubernetes; numerical methods, PDEs, optimization; physics and materials science foundations.

Contact

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Education

University of California Berkeley
Ph.D., Mechanical Engineering;
Expected: May 2026
M.S., Mechanical Engineering, 2024
M.S., Materials Science & Engineering, 2022

Harvey Mudd College
B.S. Physics, B.S. Chemistry, 2018

Management & Program Experience

- **Program Manager & People Manager**, Sakuu Corp. *May 2018 - June 2022*
  - Line managed a 10-person team of technicians and associate engineers for 3 years; partnered with staff engineers to translate research strategy into daily execution; hired, onboarded, and trained all technical staff as company scaled from 10 to 60 people.
  - Led a $300K, 1.5-year program to co-develop a browser-based ML and analytics platform with an external software partner, managing stakeholders across two companies and serving as product owner for UI/UX and requirements.
  - Defined scope, success metrics, and 3-year roadmap for a new modeling & simulation business unit; owned vendor selection, SOWs, and alignment across engineering, operations, and leadership.
  - Navigated pre-product-market-fit R&D uncertainty: balanced exploratory research with investor milestone commitments, spread risk across subteams, and kept team productive and motivated through constant change.
- **Technical Lead, Design for Manufacturing**, Apple Inc. *February 2025 - August 2025*
  - Owned end-to-end development of an internal AI-enabled platform for component-level data ingestion and visualization; coordinated ML modeling, backend infrastructure, and deployment.
  - Led evaluation of 18 ML approaches for solder joint failure prediction, framing tradeoffs between accuracy, interpretability, and compute cost; proposed a novel hybrid FEA + neural-network method and roadmapped its development.
- **Teaching Assistant, Communications for Engineering Leaders**, Fung Institute, UC Berkeley *2021 - 2025*
  - Mentored dozens of graduate students 1:1 over 4 years, quickly understanding their technical work (across diverse engineering fields) and coaching them to communicate complex ideas clearly through written, oral, and visual modes.
  - Provided constructive feedback on technical communication, framing weaknesses as areas for growth and helping students develop both skills and confidence.
- **Program Manager & Technical Lead**, UC Berkeley *June 2022 - Present*
  - **DEWA Program:** Owned execution of an international engineering certificate program; coordinated faculty, registrars, TAs, and DEWA leadership across time zones, managing schedules, enrollment, and escalations.
  - **Capstone Connect:** Led design and delivery of an automated matching platform for hundreds of industry-sponsored MEng capstone projects, translating multi-stakeholder requirements into system logic.

Research Experience

- **Graduate Student Researcher**, UC Berkeley *May 2021 - Present*
  - Led development of ML-driven optimization and simulation tools across materials discovery (The Materials Project) and agricultural systems (AI Institute for Next-Gen Food Systems), spanning ODE models, ray tracing, and experimentation pipelines.
  - Experience translating research prototypes into usable internal tools and platforms (APIs, AWS, testing, performance optimization in Python).
  - Outstanding Graduate Student Instructor 2024, Modeling and Simulation of Advanced Manufacturing Systems.